

PROFESSIONAL SUMMARY

- Obtained 2 year of experience in automotive field by working, analyzing and testing on real-time ECU's.
- Knowledge and expertise in test environment, strategies, planning, methodologies for testing the ECU software development.
- Received industrial experience in testing and analyzing with Vector CAN tools - CANalyzer, and CANape.
- Acquired basic Knowledge in Hardware in the Loop (HIL) testing with DSpace Control Desk.
- Familiar with MATLAB, Simulink, Simspace for power electronics and control systems implementation.
- Experience in developing, programming, and testing embedded software systems using Raspberry Pi, Arduino Nano, DC Motors, and sensors.
- Proficient in working with Oscilloscopes, Signal Generators, Microcontrollers and Automotive protocols.
- Knowledge communication bus protocol in CAN, J1939, Modbus, TCP/IP, LIN and I2C.
- Proficient in MS Office suite, including Outlook, for efficient communication, organization, and productivity.
- Knowledge in Electric drives, Power electronics and Control systems.
- Strong willingness to learn new technologies related to Control Systems and Embedded Software Applications used in Automotive industries.

TECHNICALSKILLS

Testing & Validation: HIL/SIL testing, Manual & Automated Testing, Bench Testing, Vehicle Testing.

Automation & Scripting: Python scripting for ECU test automation.

Communication Protocols: J1939, CAN, CAN FD, CANape.

Tools & Environments: ETAS INCA, ALERIO, Vector Tools, ECU-Test, DBC, EXE, ADO, Rally.

Software Development Process: Agile (Scrum), Test Case Management, Defect Lifecycle Management.

Test Documentation: Test Case Creation, Validation Reports, Test Plan Development.

PROFESSIONAL EXPERIENCE

Techno Square/Cube Hub

Client: John Deere

October 2023 – Present

Horicon, WI

Embedded Software Test Engineer

October 2023 – Present

- Developed and executed test cases based on Software Requirement Documents (SRD) in Excel for HIL and SIL environments.
- Conducted manual and automated testing of ECU features, creating test automation packages using Python scripting.
- Diagnosed faults by generating and analyzing DTC codes in Canalyzer.
- Performed bench testing for software validation and regression testing.
- Reprogrammed controllers from old versions to new version payloads, ensuring seamless transitions.

- Created test plans in ADO for multiple vehicle variants and executed test strategies for utility vehicles.
- Simulated nodes for accurate input/output message validation.
- Collected test data, analyzed test results, and identified defects for performance improvement.
- Collaborated with design and development teams to troubleshoot and resolve software and hardware issues.
- Applied Black-Box and White-Box testing methodologies.
- Maintained detailed test records, including test processes, results, and findings.
- Worked with CAN-based interfaces and J1939 protocols for HIL system validation.
- Generated test reports and validation summary reports at the end of each testing cycle.
- Modified test cases dynamically based on analysis and results for improved validation efficiency.
- Attended Agile meetings, actively contributing to software testing discussions and user story documentation.
- Managed defect lifecycle and triaged issues using J1939 and CAN protocols, coordinating with OEM partners for resolution.
- Writing requirements for BMS and testing battery conditions. Checking CAN messages for discharge power, charge power, voltage and current.
- Developed test plans and strategies to validate hardware components against design specifications and requirements.
- Designed and executed test cases to evaluate the performance, functionality, and reliability of hardware components, including circuit boards and electronic devices.
- Collaborated with design and development teams to resolve problems occurred during diagnose and while debugging the hardware and software issues.
- Knowledge of Jenkins, CI/CD and BOM change management.

Alphasmart Solutions
Validation Engineer

June 2023 – October 2023
Cummin, GA

- Conducted comprehensive testing and validation of electronic control units (ECU) and vehicle dynamics to ensure compliance with design specifications and safety standards.
- Automated test scripts and scenarios to enhance testing efficiency and coverage.
- Created detailed test plans and procedures based on system requirements and specifications.
- Analyzed test data and provided detailed reports on system performance and reliability.
- Coordinated with multi-disciplinary teams to integrate HIL systems with actual vehicle system.
- Ensured seamless communication and interoperability between different vehicle subsystems.
- Maintained comprehensive documentation of test cases, results and system configurations.
- Contributed to the development and refinement of testing methodologies and tools.

Assistance Engineering Trainee, Internship
Telangana Power Generation Corporate Ltd

June 2019 – December 2019
Telangana, India

- Monitored and controlled activities associated with hydropower generation.
- Operated plant equipment, such as turbines, pumps, valves, gates, fans, electric control boards, and battery banks.
- Monitored equipment operation and performance, making necessary adjustments and repairs for optimal performance.
- Monitoring the load dispatch and the time synchronization of the machine to the grid.
- Collaborated with cross-functional teams, including mechanical engineers, technicians, and operators.

Diploma Apprentice***Defense Research and Development Laboratory******November 2014 – May 2015******Telangana, India***

- Assisted in the installation and commissioning of CNC machines, ensuring proper electrical connections and functionality.
- Supported electrical engineers in troubleshooting electrical and control system issues.
- Assisted in the maintenance and calibration of motor drives, servo systems, and feedback devices.
- Conducted electrical inspections and preventive maintenance activities to ensure optimal machine performance.
- Effectively communicated and collaborated with other teams.

EDUCATION**Master of Science in Electrical Engineering*****Gannon University******August 2021— May 2023******Erie, PA*****Bachelor of Technology in Electrical & Electronics Engineering*****Jawaharlal Nehru Technological University******July 2017—November 2020******Hyderabad, India***